

- ▶ Document technical design specs in technical design document.
- ▶ Design, maintain, implement, test and debug AI code.
- ▶ Development and maintenance of reusable advanced AI tools, features and pipelines.
- ▶ Find and fix AI bugs flagged by the Quality Control team.

Must-have skills for an AI Programmer

- ▶ Demonstrated knowledge of good software engineering practices.
- ▶ Understanding of memory management, multiple processor use, and runtime optimization.
- ▶ Strong C/C++ programming skills
- ▶ Strong Object Oriented design skills
- ▶ Strong 3D Math, Trigonometry, Calculus and Linear Algebra skills
- ▶ Experience in self/college project: Path finding, locomotion, collision detection, finite state machines, game logic etc
- ▶ Experienced in implementing character AI systems in real-time games.
- ▶ Knowledge of Neural Networks, Genetic Algorithms, Fuzzy State Logic, and A-Life.

Game Tools Programmer

“Right now we have talent and tools, but soon we’ll have talented tools. New intelligent software will help game developers speed up the process as never before.”

-Duane Alan Hahn (2005)

The tools/engine programmer writes tools that can handle game-specific tasks. These tools may/may not be included in the game but help the game development process accelerate. A bad tool can hamper the game dev process where as a good tool can make it a piece of cake. These tools include scripting tools, model viewer, map editors, level editors, dynamic map generators, and custom tools for a specific game. They help the artists and game designers interface with the game core engine.

What does a Tools Programmer do?

- ▶ Program tools that help designers, artists etc to make their work easier.
- ▶ Document the tool feature to help artists and programmer utilize them fully.
- ▶ Participate in the design and implementation of games.
- ▶ Collaborate with Producers, Designers, third parties and other technical team members to plan and implement tools and pipelines.